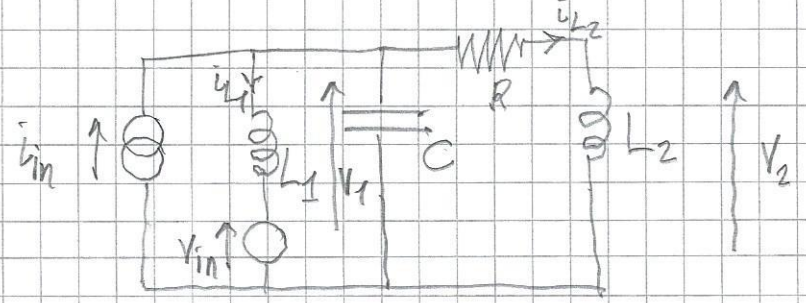
Solution Problem 1-1 (see also m-file)



SSC
$$\frac{d\vec{x}}{dt} = A\vec{x}(t) + B\vec{u}(t)$$
$$y(t) = C\vec{x}(t) + D\vec{u}(t)$$

A = 3×3
B = 3×2
C = 1×3
D = 1×2

where
$$\vec{x}(t) = \begin{bmatrix} i_{L_1}(t) \\ V_1(t) \\ i_{L_2}(t) \end{bmatrix} \quad \vec{u}(t) = \begin{bmatrix} i_{in}(t) \\ V_{in}(t) \end{bmatrix} \quad y(t) = V_2(t)$$

- $$H_1(s) = \frac{V_2(s)}{I_{in}(s)} = sL_2 \frac{sL_1 \parallel \frac{1}{sC}}{sL_1 \parallel \frac{1}{sC} + R + sL_2} = \dots$$
$$= \frac{\frac{1}{C}s^2}{s^3 + \frac{R}{L_2}s^2 + \frac{1}{C}\left(\frac{1}{L_1}+\frac{1}{L_2}\right)s + \frac{R}{L_1L_2C}}$$

- $$H_2(s) = \frac{V_2(s)}{V_{in}(s)} = \frac{sL_2}{R+sL_2} \frac{(R+sL_2) \parallel \frac{1}{sC}}{sL_1 + (R+sL_2) \parallel \frac{1}{sC}} = \dots$$
$$= \frac{\frac{1}{L_1C}s}{s^3 + \frac{R}{L_2}s^2 + \frac{1}{C}\left(\frac{1}{L_1}+\frac{1}{L_2}\right)s + \frac{R}{L_1L_2C}}$$

- SSC
$$\begin{cases} V_1 - V_{in} = L_1 \frac{di_{L_1}}{dt} \Rightarrow \frac{di_{L_1}}{dt} = \frac{1}{L_1}V_1 - \frac{1}{L_1}V_{in} \\ i_{in} - i_{L_1} - i_{L_2} = C\frac{dV_1}{dt} \Rightarrow \frac{dV_1}{dt} = \frac{1}{C}(i_{in} - i_{L_1} - i_{L_2}) \\ L_2\frac{di_{L_2}}{dt} = V_1 - R\,i_{L_2} \Rightarrow \frac{di_{L_2}}{dt} = \frac{1}{L_2}V_1 - \frac{R}{L_2}i_{L_2} \\ V_2 = V_1 - R\,i_{L_2} \end{cases}$$

$$\Downarrow$$

$$\begin{bmatrix} \frac{di_{L_1}}{dt} \\ \frac{dV_1}{dt} \\ \frac{di_{L_2}}{dt} \end{bmatrix} = \begin{bmatrix} 0 & 1/L_1 & 0 \\ -1/C & 0 & -1/C \\ 0 & 1/L_2 & -\frac{R}{L_2} \end{bmatrix} \begin{bmatrix} i_{L_1}(t) \\ V_1(t) \\ i_{L_2}(t) \end{bmatrix} + \begin{bmatrix} 0 & -\frac{1}{L_1} \\ \frac{1}{C} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} i_{in}(t) \\ V_{in}(t) \end{bmatrix}$$

$$V_2(t) = \begin{bmatrix} 0 & 1 & -R \end{bmatrix} \begin{bmatrix} i_{L_1}(t) \\ V_1(t) \\ i_{L_2}(t) \end{bmatrix} + \begin{bmatrix} 0 & 0 \end{bmatrix} \begin{bmatrix} i_{in}(t) \\ V_{in}(t) \end{bmatrix}$$

- integrator $H(s) = \frac{1}{s}$

SCC $\frac{dx}{dt} = 0 \cdot x(t) + 1 \cdot u(t)$

$y(t) = 1 \cdot x(t)$

Interpretation: $\frac{dx}{dt} = u(t)$

$\Downarrow$

$y(t) = x(t) = \int u(t)dt$ QED.

- differentiator $H(s) = s$

Implicit SSC $E\frac{d\vec{x}}{dt} = A\vec{x}(t) + Bu(t)$

$y(t) = C\vec{x}(t) + Du(t)$

Here $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} \frac{dx_1}{dt} \\ \frac{dx_2}{dt} \end{bmatrix} = I\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \end{bmatrix}u(t)$

$y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix}\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$

Interpretation $\begin{cases} \frac{dx_2}{dt} = x_1(t) \\ 0 = x_2(t) - u(t) \end{cases}$

$\Downarrow$

$\begin{cases} x_2(t) = u(t) \\ x_1(t) = \frac{du}{dt} \end{cases}$

$y(t) = x_1(t) = \frac{du}{dt}$ QED.

Solution Problem 1-2 (see also m-file)

- $H(z) = \dfrac{b_0 + b_1 z^{-1} + b_2 z^{-2} + b_3 z^{-3}}{1 + a_1 z^{-1} + a_2 z^{-2} + a_3 z^{-3}}$

- $H(z) \rightarrow$ SSD by hand:

$$H(z) = \frac{Y(z)}{U(z)} = \frac{b_0 z^3 + b_1 z^2 + b_2 z^1 + b_3}{z^3 + a_1 z^2 + a_2 z^1 + a_3} \cdot \frac{V(z)}{V(z)}$$

$$\Downarrow \quad \begin{cases} U(z) = (z^3 + a_1 z^2 + a_2 z + a_3) V(z) \\ Y(z) = (b_0 z^3 + b_1 z^2 + b_2 z + b_3) V(z) \end{cases}$$

$$\Downarrow \quad \begin{cases} u(n) = v(n+3) + a_1 v(n+2) + a_2 v(n+1) + a_3 v(n) \\ y(n) = b_0 v(n+3) + b_1 v(n+2) + b_2 v(n+1) + b_3 v(n) \end{cases}$$

Choose $\vec{x}(n) = \begin{bmatrix} x_1(n) \\ x_2(n) \\ x_3(n) \end{bmatrix} = \begin{bmatrix} v(n) \\ v(n+1) \\ v(n+2) \end{bmatrix}$

Then
$$\begin{aligned} x_1(n+1) &= x_2(n) \\ x_2(n+1) &= x_3(n) \\ x_3(n+1) &= v(n+3) \\ &= u(n) - a_1 v(n+2) - a_2 v(n+1) - a_3 v(n) \end{aligned}$$

And
$$\begin{aligned} y(n) &= b_0 \cdot v(n+3) + b_1 v(n+2) + b_2 v(n+1) + b_3 v(n) \\ &= b_0 \left[u(n) - a_1 v(n+2) - a_2 v(n+1) - a_3 v(n) \right] \\ &\quad + b_1 v(n+2) + b_2 v(n+1) + b_3 v(n) \end{aligned}$$

SSD
$$\vec{x}(n+1) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_3 & -a_2 & -a_1 \end{bmatrix} \vec{x}(n) + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(n)$$

$$y(n) = \begin{bmatrix} b_3 - b_0 a_3 & b_2 - b_0 a_2 & b_1 - b_0 a_1 \end{bmatrix} \vec{x}(n) + b_0 \cdot u(n)$$

- integrator $H(z) = \dfrac{1}{1 - z^{-1}}$ SSD
$$\begin{aligned} x(n+1) &= x(n) + u(n) \\ y(n) &= x(n) + u(n) \\ &= \sum_{i=0}^{\infty} u(n+i) \end{aligned}$$
QED

- differentiator $H(z) = 1 - z^{-1}$ SSD
$$\begin{aligned} x(n+1) &= 0 \cdot x(n) + u(n) \\ y(n) &= -x(n) + u(n) \\ &= u(n) - u(n-1) \end{aligned}$$
QED